

Program : Diploma in Renewable Energy	
Course Code : 4297	Course Title: ELECTRICAL MACHINES LAB 1
Semester : 4	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To Understand the characteristics of dc & compound generator
- To get familiar the speed characteristics of dc shunt motor
- To perform different test on DC machine
- To impart basic knowledge on efficiency & regulation of single phase transformers

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge of fundamentals of electrical.		Fundamentals of Electrical & Electronics Engineering	4

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Plot different characteristics of dc generators and perform load test	12	Applying
CO2	Identify various methods of starting and speed control of a dc motor	9	Applying
CO3	Make use of standard tests to determine the efficiency of a DC machine	9	Applying
CO4	Utilize various tests to determine the efficiency and voltage regulation of a single phase transformer	9	Applying

	Lab Test	6	
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CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Plot different characteristics of dc generators and perform load test		
M1.01	Identify different parts of dc machine	3	Understanding
M1.02	Plot magnetization characteristics of a d.c. shunt generator	3	Applying
M1.03	Perform load on d.c. shunt generator	3	Applying
M1.04	Predetermine the efficiency of a dc shunt machine using Swinburn's test	3	Applying
CO2	Identify various methods of starting and speed control of a dc motor		
M2.01	Start DC shunt motor by using three point starter & four point starter	3	Applying
M2.02	Perform load test on DC shunt motor and plot the mechanical characteristics.	3	Applying

M2.03	Control the speed of dc shunt motor by (a) Armature resistance control (b) Field control	3	Applying
	Lab Test – I	3	
CO3	Make use of standard tests to determine the efficiency of a DC machine		
M3.01	Separate the losses of a dc shunt motor by conducting Swinburns test	3	Applying
M3.02	Plot the mechanical characteristics of a dc series motor .	3	Applying
M3.03	Determine the efficiency of a DC machine by conducting Hopkinson's test	3	Applying
C04	Utilize various tests to determine the efficiency and voltage regulation of a single phase transformer		
M4.01	Obtain efficiency and voltage regulation of a single phase transformer by OC & SC tests	3	Applying
M4.02	Perform OC & SC tests on single phase transformer and draw equivalent circuit	3	Applying
M4.03	Determine the efficiency of a single phase Transformer by Direct loading	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Bimbura P. S., Electrical Machinery, 7/e, Khanna Publishers, 2011
T2	Theraja B. L., A Textbook of Electrical Technology, S. Chand & Company, New Delhi

R3	J B Gupta, “Theory and Performance of Electrical Machines”, S K Kataria & Sons publications, 2013
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Online Resources

Sl.No	Website Link
1	https://www.ee.iitb.ac.in/course/~emlab/lab-manual.html
2	Virtual labs: http://vlabs.iitb.ac.in/vlab/labsee.html
3	http://em-coep.vlabs.ac.in
4	http://vem-iitg.vlabs.ac.in
5	www.swayam.gov.in
6	www.nptel.ac.in
7	https://www.electrical4u.com/electric-machines/
8	www.youtube.com
9	https://www.pscmr.ac.in/eee_labmanuals/2-2_ELECTRICAL%20MACHINES-I%20LAB.pdf
10	http://vitengineering.com/lab/EE6411-ELECTRICAL-MACHINES-LABORATORY-I.pdf
11	https://1filedownload.com/wp-content/uploads/2019/08/Electric-Machines-By-D.P.Kothari.I.Nagrath.pdf
12	https://nptel.ac.in/courses/108/105/108105017/

SAMPLE LAB QUESTIONS

1. Predetermine the efficiency of the two identical shunt machines are coupled mechanically and also connected electrically in parallel, Plot the efficiency versus load power characteristics of both the machines.
2. Conduct the load test on the given static machines which consists of two electric circuits interlinked by a common magnetic circuit and plot the efficiency versus load power characteristics.
3. Predetermine the efficiency of the machine transfers energy from one circuit to another without a change in frequency. Plot the efficiency and regulation characteristics.
4. Conduct a suitable experiment to predetermine the efficiency characteristics of the DC machine when working as a generator and motor excluding the stray losses.